

Decoding Interfacial Evolution of Aluminum Anode and Constructing Multifunctional Layers toward Ultra-Long Cycle Stability

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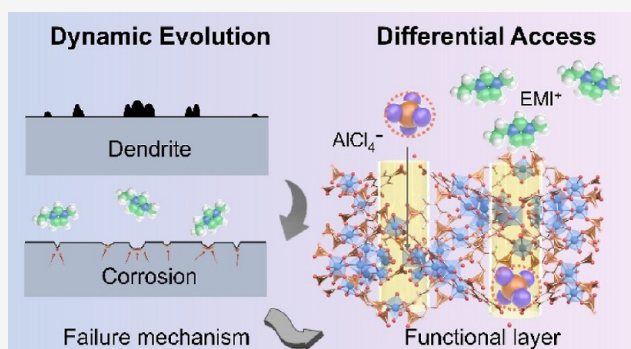


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ABSTRACT: Understanding the dynamic evolution of the aluminum anode interface in ionic liquid electrolytes is crucial for the large-scale application of rechargeable aluminum batteries (RABs). Herein, we use a series of advanced in situ characterizations to reveal a dendrite-to-corrosion transition, demonstrating that imidazolium cations (EMI^+) fundamentally dominate anode degradation and instability. Therefore, based on a differential access mechanism, we engineer a metal–organic framework (MOF-C) layer with molecular-scale nanochannels that selectively block corrosive EMI^+ while accelerating AlCl_4^- diffusion. The modified Al/MOF-C anode achieves an unprecedented cycling stability of >11,000 h (at 1 mA cm^{-2} with minimal 20 mV overpotential) in symmetric cells, far exceeding all previous reports (typically <2000 h). Matched with natural graphite cathodes, full cells retain 95% capacity over 500 cycles. This work not only resolves the long-standing interfacial dispute in RABs but also establishes an ingenious solution aligned with interfacial evolution for next-generation stable metal anodes.



INTRODUCTION

Against the backdrop of the global energy transition and the urgent imperative to handle climate change, the development of efficient, cost-effective, and sustainable energy storage technologies has emerged as a pivotal issue for achieving carbon neutrality goals.¹ Although lithium-ion batteries have long dominated the energy storage market due to their high energy density and well-established technology systems, the scarcity of lithium resources and the sharp price fluctuations have impelled the academic community to explore alternative energy storage systems.^{2,3} Among them, rechargeable aluminum batteries (RABs) stand out because Al, the most abundant metal element in the Earth's crust, has the characteristics of low cost, high theoretical capacity (2980 mAh g^{-1}), and environmental friendliness.^{4,5} Moreover, the theoretical energy density of RABs can reach $150\text{--}200 \text{ Wh kg}^{-1}$.^{6–12} However, the practical application of RABs is still limited by short cycle life and low anode utilization, attributed to the severe interfacial failure problems of Al anodes.^{13,14} It is indicated that overcoming the stability bottleneck of Al anodes is the key to unlocking the large-scale application of RABs.

To enhance the stability of Al anodes, researchers have proposed various modification methods: (1) artificial solid–electrolyte interphases (SEIs), such as Al_2O_3 , AlF_3 , etc., are designed to physically shield the anode, thereby reducing

undesirable side reactions with the electrolyte;^{15–17} (2) alloying strategies, including Al–Mg, Al–Cu, etc., introduce heterogeneous elements to modulate the nucleation overpotential of Al, improving the deposition behavior;^{18–20} (3) electrolyte optimization, through approaches like additive design, eutectic solvent design, etc., aims to stabilize the electrochemical environment and suppress side reactions.^{21–25} However, the above strategies fail to achieve the Al anode with stable long-term cycling, as the cycle life of its symmetric cells is typically less than 2000 h under practical conditions.^{26,27} The fundamental reason lies in the fact that existing methods fail to effectively address the interfacial instability caused by dynamic changes of Al anodes during cycling. Specifically, artificial SEIs crack due to volume changes during Al plating/stripping processes, elemental segregation easily occurs at the alloying interface, and additives may introduce new side reaction pathways.²⁸ These limitations highlight the deficient

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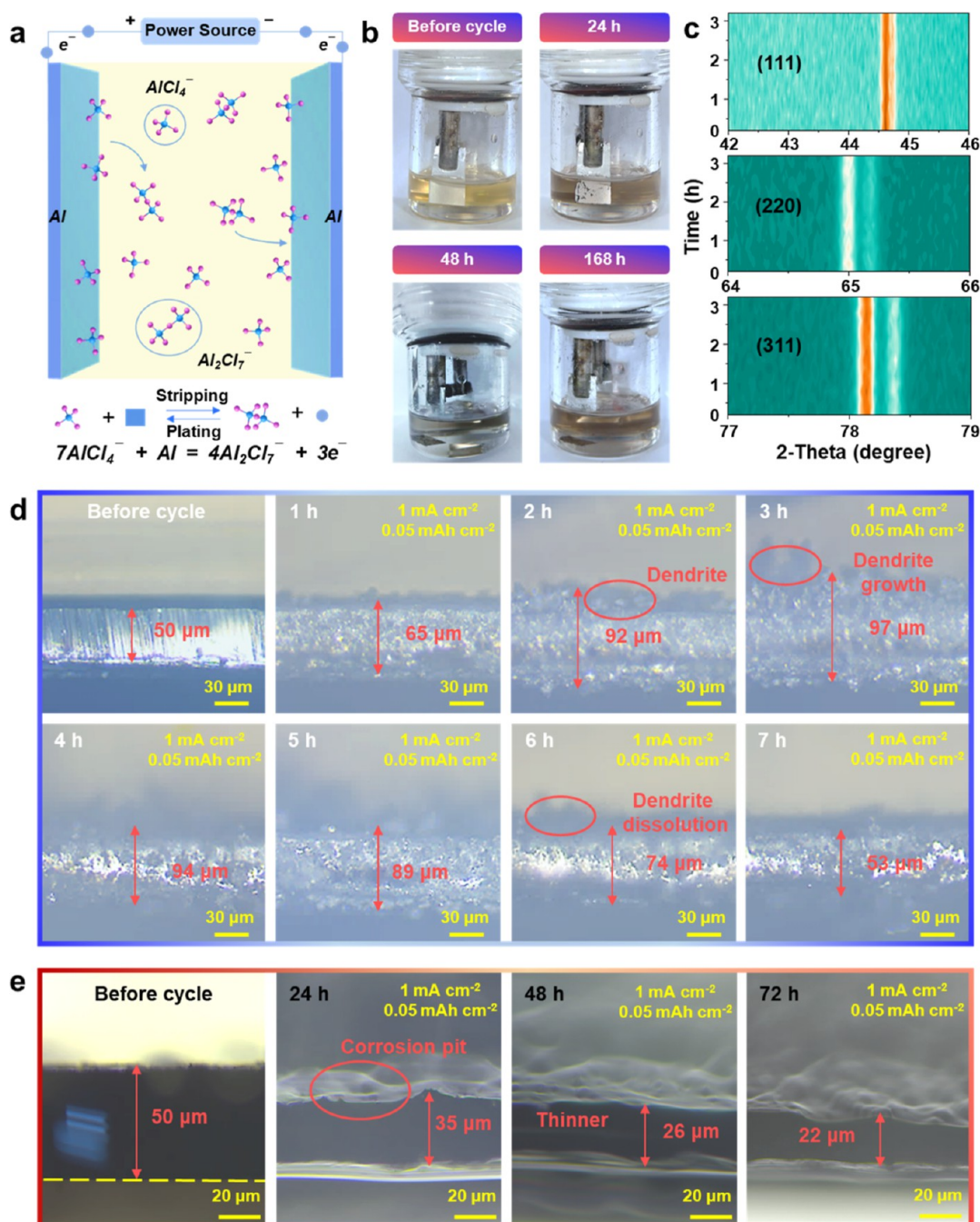


Figure 1. (a) Schematic diagrams of the plating/stripping mechanism. (b) In situ optical images of the Al metal surface in the Al–Al symmetrical cell during actual operation under testing conditions of 1 mA cm^{-2} and 1 mAh cm^{-2} . (c) In situ XRD patterns of the Al anode. (d) In situ optical microscopy images of the Al metal surface in the Al–Al symmetrical cell (1 mA cm^{-2} and 0.05 mAh cm^{-2}). (e) In situ optical microscopy images of the Al metal surface in the Al–Al symmetrical cell after long-term cycling (1 mA cm^{-2} and 0.05 mAh cm^{-2}).

cies of traditional protection strategies, necessitating the establishment of mechanism-driven new approaches through the systematic elucidation of interfacial evolution mechanisms.

Currently, the controversy over the failure mechanisms of Al anodes mainly centers on two core issues: dendrite growth and surface corrosion. Some studies observed dendrites on the Al surface after cycling and thus concluded that dendrite growth is the primary cause of Al anode failure.^{29–33} In contrast, other studies argued that the continuous corrosion of Al by the electrolyte leads to the irreversible loss of Al, which in turn causes the failure of the Al anode.^{34–36} It can be considered that the controversy stems from the fact that traditional

characterization methods can only provide static interface information and have difficulty capturing the dynamic evolution of the interface structure during the charging–discharging processes.³⁷ Moreover, the interaction mechanisms between various ions in ionic liquids and the Al anode are still unclear, hindering the precise identification of the root cause of Al anode failure. Therefore, developing advanced in situ techniques to gain a deeper understanding of the interaction mechanisms between ions and Al anode is crucial for capturing the dynamic evolution of interface and elucidating the underlying causes of Al anode failure.

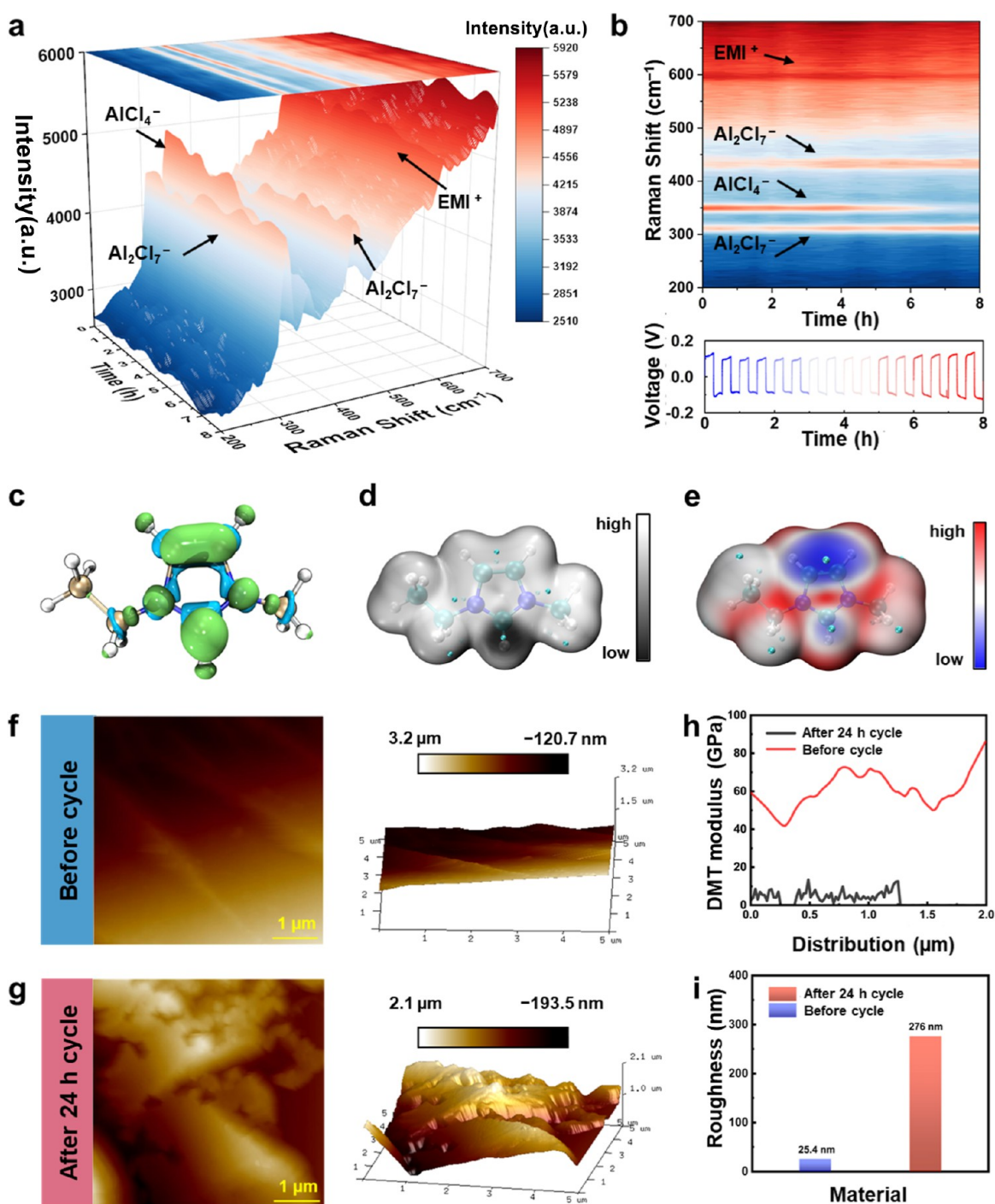


Figure 2. (a) Three-dimensional in situ Raman spectra of the ionic liquid after different cycling times. (b) Contour plot of the in situ Raman spectra. (c) Electrostatic potential distribution of the EMI^+ ion. (d) Local electron attachment energy distribution of the EMI^+ ion. (e) Local ionization energy distribution of the EMI^+ ion. (f) AFM image of the Al foil before cycling. (g) AFM image of the Al foil after 24 h of cycling (1 mA cm^{-2} , 1 mAh cm^{-2}). (h) Comparison of DMT measurements of the Al foil before and after 24 h of cycling. (i) Comparison of surface roughness of the Al foil before and after 24 h of cycling.

Herein, for the first time, the dynamic evolution of the Al anode interface in ionic liquids is clarified by combining in situ optical microscopy and in situ X-ray diffraction (XRD). The results indicate that during the initial cycles, dendrites grow on the Al surface. As cycling progresses, the dendrites gradually disappear, accompanied by the formation of corrosion pits. Ultimately, the loss of active Al material caused by corrosion becomes the dominant factor in Al anode failure. Importantly, through in situ Raman spectroscopy, we unequivocally confirm

that the EMI^+ cation plays a critical role in inducing Al corrosion, providing direct experimental evidence for the long-debated corrosion mechanism of Al anodes in ionic liquids. Based on this phenomenon, we propose an ingenious strategy of utilizing metal–organic frameworks (MOFs) with tunable pore structures and high chemical stability to modify the Al metal anode.³⁸ The MOF protective layer (Al/MOF-C) can be grown on the Al surface via a hydrothermal method. The nanochannels of the MOF-C layer with the differential access

mechanism precisely block EMI^+ through size effects while accelerating the transport of AlCl_4^- , thereby achieving dual functions of corrosion inhibition and mass transport promotion. Electrochemical tests demonstrate that the Al/MOF-C-symmetric cell exhibits an unprecedented cycling stability of 11,000 h at 1 mA cm^{-2} with an overpotential of only 20 mV, surpassing all previously reported Al-based symmetric cells. The full cell paired with a flake graphite cathode (FG//Al) maintains a high specific capacity of 120 mAh g^{-1} , with a capacity retention rate of 95% after 500 cycles. Through dynamic interfacial evolution analysis and MOF differential access mechanism design, this study provides significant theoretical and experimental support for the technological development and practical application of RABs.

RESULTS AND DISCUSSION

Interface Evolution of Al Metal in Ionic Liquid Systems

Figures 1a and S1 illustrate the schematic diagrams of the Al plating and stripping mechanisms in an Al–Al symmetrical battery, where Figure S1 presents the corresponding galvanostatic charge–discharge curves of the Al–Al symmetrical cell under ideal operating conditions. During the Al plating and stripping processes, AlCl_4^- and Al_2Cl_7^- ions move between the Al electrodes, as described by the following reactions:³⁹



The corresponding charge–discharge curves demonstrate the stable cycling performance, where the overpotentials for Al plating and stripping also remain relatively stable without significant fluctuations. Figure 1b presents in situ optical images of the Al metal surface taken at different intervals during the cycling of an Al–Al symmetrical battery. Initially, the Al surface is smooth and flat. However, as the cycling time increases, corrosion pits gradually appear on the Al surface. These pits become more numerous over time, eventually leading to the breaking and dissolution of the Al foil into the ionic liquid, with the electrolyte color changing from pale yellow to yellow-brown. Figure S2 shows optical images of the ionic liquid electrolyte after being left in the cell for the same duration without cycling, and the electrolyte color remains unchanged. These phenomena macroscopically indicate severe corrosion of the Al sheets during cycling. Figure 1c and S3 show the in situ XRD patterns of the Al surface in the symmetrical battery. The diffraction peaks at 44.6° , 64.95° , and 78.2° for the original Al correspond to the (111), (220), and (331) planes, respectively.²⁸ The intensities of the Al diffraction peaks gradually weaken with an increased cycling time, indicating the gradual dissolution of Al during cycling and further corroborating Al corrosion.

Figure 1d provides detailed in situ optical microscopy images of the Al surface under a current density of 1 mA cm^{-2} and a deposition capacity of 0.05 mAh cm^{-2} , showing the interfacial evolution at different cycling times. At the beginning of cycling, the Al surface is uniform. As cycling progresses, dendrites start to form and thicken. Within the initial 3 h, the thickness of the dendrite layer significantly increases, stabilizing between 3 and 4 h, at which point the thickness of the Al sheet is $97 \text{ }\mu\text{m}$. After 4 h, the thickness begins to decrease gradually, and after 7 h of cycling, the thickness of the Al sheet is $53 \text{ }\mu\text{m}$. It is suggested that dendrite growth occurs on the Al metal surface in the short term but gradually

dissolves with an increased cycling time (after 4 h of cycling). To elucidate the dendrite dissolution process, scanning electron microscopy (SEM) tests were conducted on Al surfaces cycled under the same conditions for the same time, as shown in Figure S4. Significant corrosion pits can be observed on the Al surface, and as the cycling time increases, the corrosion worsens, demonstrating the simultaneous presence of corrosion and dendrites. Figure 1e shows the Al surface under an optical microscope after long-term cycling, where corrosion pits are clearly visible and the Al metal layer gradually becomes thinner. The Al surface is initially smooth with a uniform thickness of $50 \text{ }\mu\text{m}$, but corrosion becomes more apparent over time. After 72 h of cycling, the Al metal significantly thins to $22 \text{ }\mu\text{m}$. This indicates that dendrite formation is a critical issue in the short term, while corrosion becomes the primary factor affecting Al electrode integrity and performance after long-term cycling as dendrites disappear. As shown in Figure S5, an in situ mold was designed in this work to observe real-time changes on the Al surface during charge and discharge. Figure S6 provides $500\times$ magnification in situ optical microscopy images of the Al surface, showing the gradual and severe formation of corrosion. Supporting Information Videos 1 and 2 illustrate the surface morphology changes during the charge–discharge processes of the Al symmetrical battery, further confirming the presence of corrosion. Experimental results indicate that both corrosion and dendrites coexist on the Al anode electrode surface, affecting the Al electrode during different cycling stages. In the initial few hours (less than 4 h), the increase in dendrite layer thickness suggests that dendrite formation dominates electrode degradation. With a further cycling time, the dendrite layer stabilizes and eventually begins to thin (4–7 h). After long-term cycling (beyond 24 h), dendrites disappear and corrosion dominates. Therefore, understanding the corrosion mechanism of Al in ionic liquid systems is crucial for improving the performance and lifespan of the RABs.

Corrosion Mechanism of Al Metal in Ionic Liquid Systems

As shown in Figure 2a,b and S7, the corrosion mechanism of Al in ionic liquids was analyzed in detail through in situ Raman spectroscopy tests in an Al symmetrical battery (1 mA cm^{-2} and 0.25 mAh cm^{-2}), with Figure S7 providing the Raman spectra of the ionic liquid collected at different cycling times. Figure 2a presents a three-dimensional variation of Raman spectral intensity with time and Raman shift, where 311 cm^{-1} and 433 cm^{-1} correspond to the dimer Al_2Cl_7^- , 350 cm^{-1} corresponds to the monomer AlCl_4^- , and 597 cm^{-1} corresponds to EMI^+ , respectively.⁴⁰ As the cycling time increases, the relative intensities of EMI^+ and AlCl_4^- gradually decrease, while the relative intensity of Al_2Cl_7^- significantly increases. Figure 2b more intuitively shows the Raman spectra at different times and the charge–discharge curves of the Al symmetrical battery. In situ Raman spectra clearly show the peaks at 311 cm^{-1} and 433 cm^{-1} becoming stronger over time, while the peaks at 350 cm^{-1} and 597 cm^{-1} gradually weaken. Based on this, the corrosion mechanism of the Al anode in the ionic liquid can be deduced. First, during the discharge process of the battery, EMI^+ absorbs electrons, and the Al electrode loses electrons to form Al^{3+} ions. Subsequently, the electron-depleted Al is combined with AlCl_4^- to form Al_2Cl_7^- . Therefore, as cycling progresses, the concentrations of EMI^+ and AlCl_4^- gradually decrease, while the concentration of Al_2Cl_7^- gradually increases, as reflected in the in situ Raman

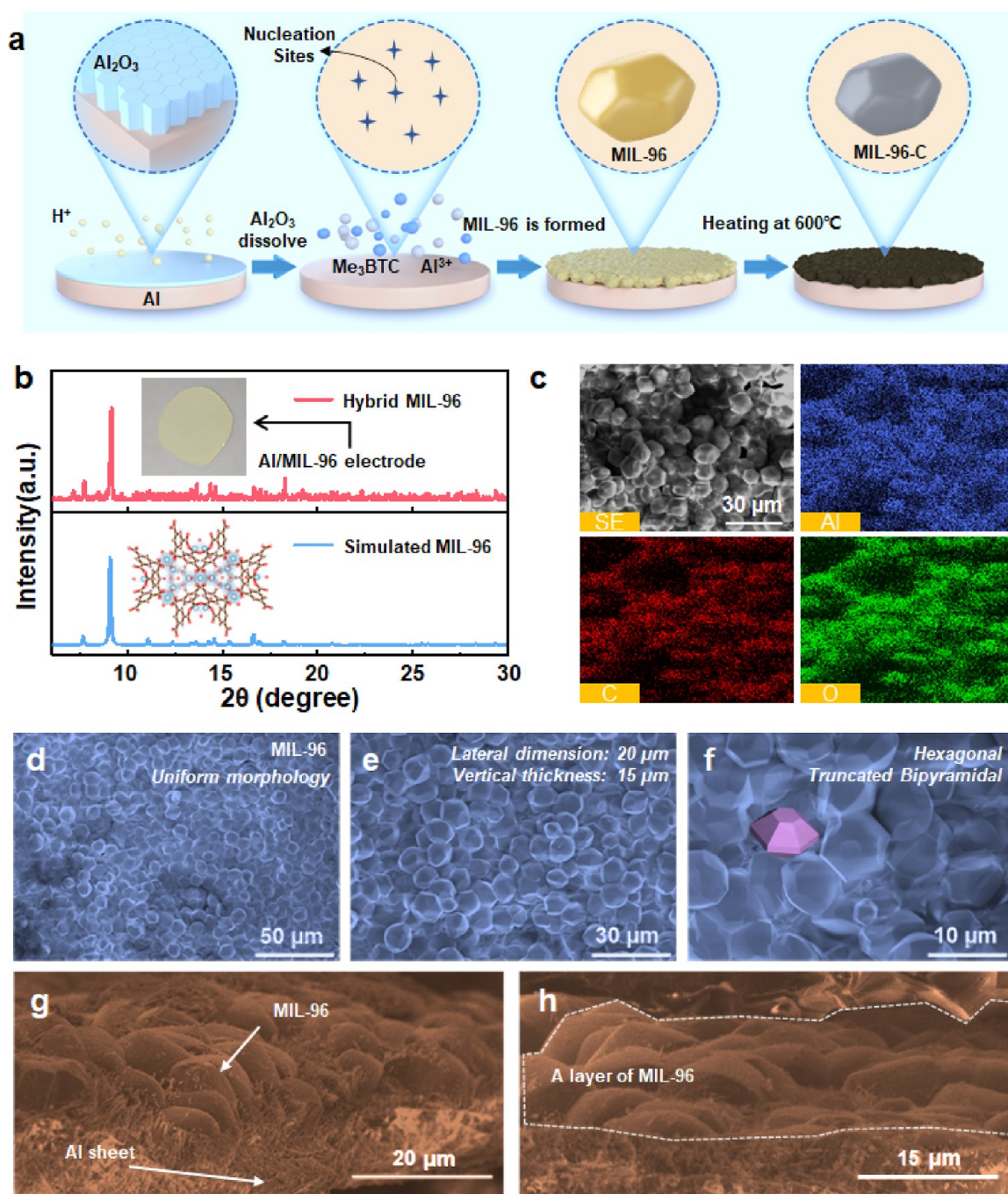


Figure 3. (a) Schematic diagram of the process of in situ construction of a MOF functional layer on the Al surface. (b) XRD pattern of the synthesized MIL-96. (c) Elemental analysis of the synthesized MIL-96. (d–f) SEM images of MIL-96. (g,h) Cross-sectional SEM images of MIL-96 on the Al surface.

spectra. To further validate the reactivity of EMI^+ on Al surfaces, we conducted density functional theory calculations (B3LYP/6-31G* level),^{41,42} which revealed the Fukui index distribution (Figure 2c and S10) and identified reactive sites on EMI^+ through electron density variations. The atomic numbering and optimized geometry of EMI^+ were determined (Figures S8 and S9) combined with Fukui index analysis (Figure 2c and S10), revealing its intrinsic electronic structure, while the electrostatic potential distribution explicitly identifies the electron-deficient region at C4 and H11 (Figure 2c). This localized electron deficiency aligns with the low Local Electron Attachment Energy (LEAE)⁴³ values in the same region

(Figure 2d), suggesting a thermodynamic preference for electron acquisition during interfacial reactions. Furthermore, the reduced Local Ionization Energy (LIE) at C4–H11 (Figure 2e) implies that electrons in these orbitals are readily lost, mechanistically supporting the observed EMI^+ -mediated oxidative corrosion process. These theoretical calculations indicate that the C4–H11 positions on the EMI^+ ions are more likely to absorb electrons from Al and react with Al, exhibiting a higher reactivity in chemical reactions. Figure 2f,g shows the AFM images of the Al sheet before and after 24 h of cycling. Before cycling, the Al surface is relatively smooth and uniform. After 24 h of cycling, the surface becomes

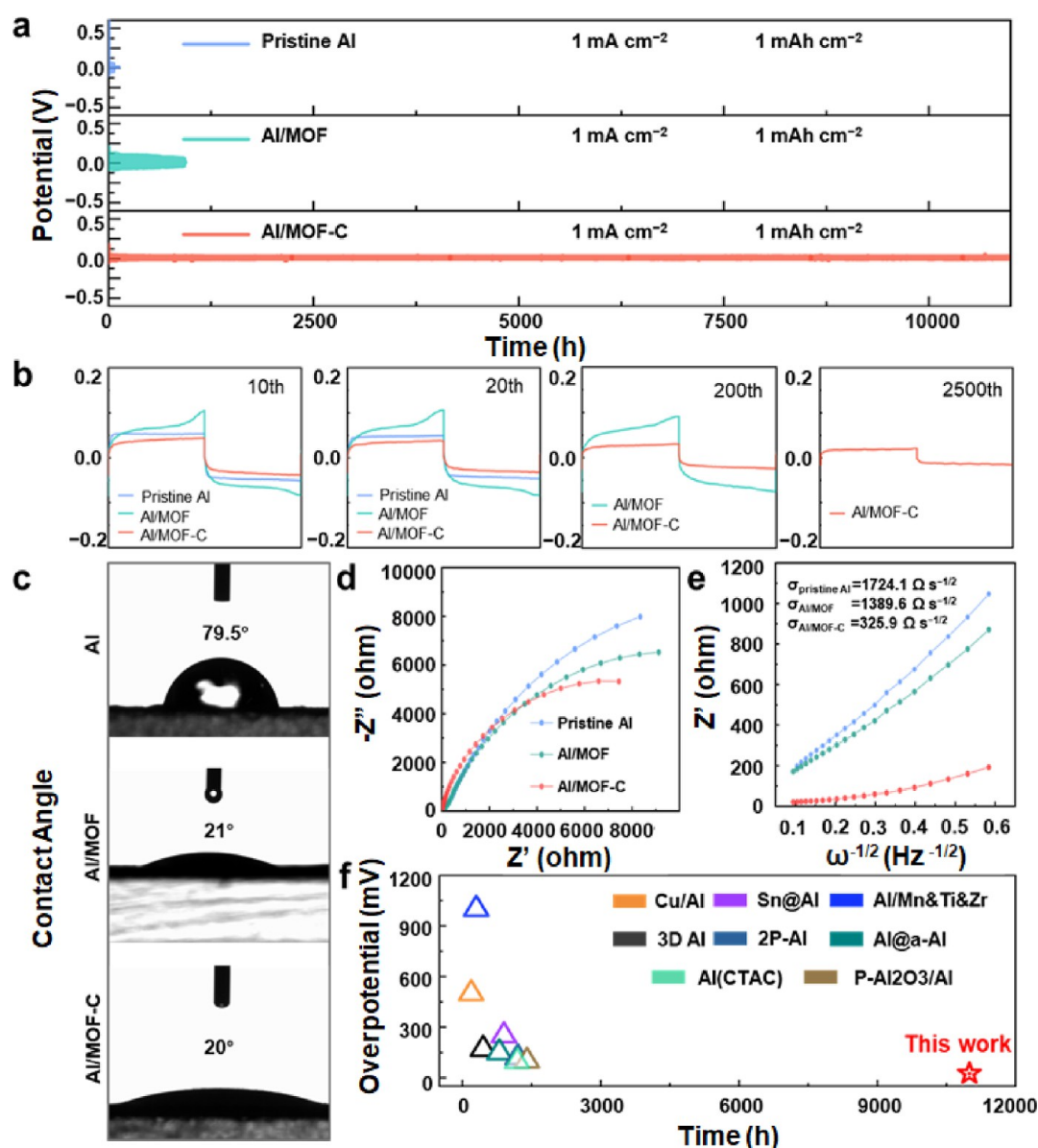


Figure 4. (a) Comparison of the cycling stability of symmetrical cells assembled with different anodes under conditions of 1 mA cm⁻² and 1 mAh cm⁻². (b) Voltage distribution of different symmetrical cells at the 10th, 20th, 200th, and 2500th cycles. (c) Contact angle measurements of different anodes with the ionic liquid. (d) AC impedance spectra of different symmetrical cells. (e) Relationship between the real part of the impedance and low frequency. (f) Comparison of cycling time and overpotential between our work and previous studies.^{45–52}

significantly rougher, with numerous pits and protrusions appearing. Figure 2h shows the comparison of mechanical properties of the Al sheet before and after cycling. After cycling, the Derjaguin–Muller–Toporov (DMT) value significantly decreases and fluctuates more, indicating reduced surface hardness and stiffness and deteriorated mechanical properties. Figure 2i shows the change in surface roughness, increasing from 25.4 nm before cycling to 276 nm after cycling. Figures S11 and S12 compare the surface morphology of the Al sheet before and after cycling for 24 h using an optical profilometer. The results are consistent with the AFM results, showing that the Al sheet becomes rougher and significantly less smooth after cycling. These results indicate that during cycling, the reaction between Al sheet and EMI⁺ ions directly leads to Al corrosion, causing the Al surface to become rougher and the mechanical properties to significantly decrease, thereby

affecting the stability of Al metal anode during long-term battery cycling.

In Situ Construction of Functionalized Interface Layers on the Al Metal Surface

To enhance the corrosion resistance of the Al metal anode, we innovatively constructed a protective layer in situ on the Al surface. As shown in Figure 3a, under hydrothermal conditions, Al₂O₃ on the Al surface dissolves and releases Al³⁺ ions, which can react with 1,3,5-benzenetricarboxylic acid (Me₃BTC) to form a MIL-96 (Al/MOF anode) protective layer through nucleating and growing on the Al surface.⁴⁴ Subsequently, after heating at 600 °C for 12 h, MIL-96 undergoes partial carbonization and transforms into a MIL-96-C (Al/MOF-C anode) protective layer. Figure 3b shows the comparison between the XRD pattern of the synthesized MIL-96 and the simulated XRD pattern, which demonstrates that synthesized MIL-96 closely matches simulated MIL-96 in

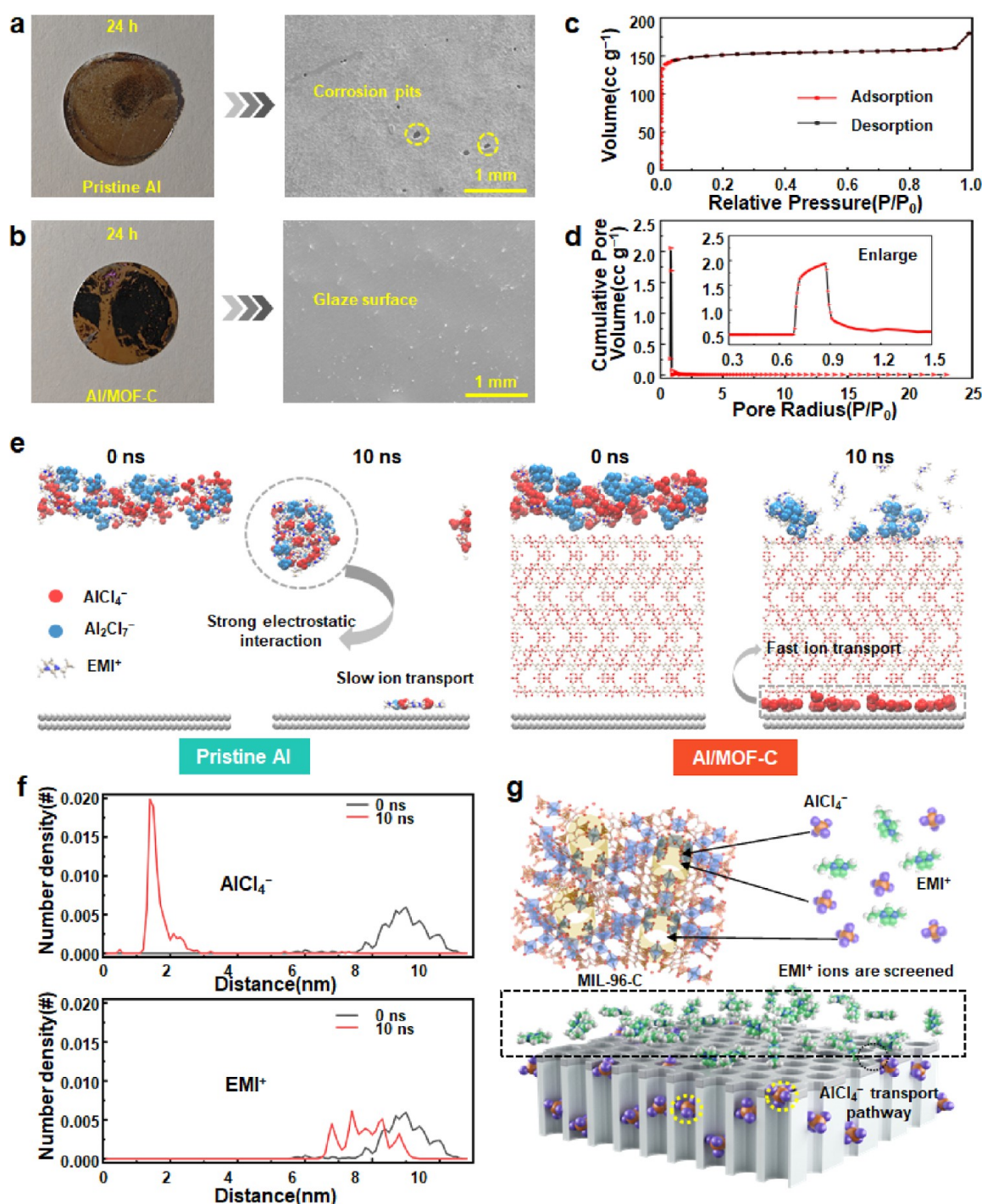


Figure 5. SEM images of the surface of different anodes after cycling for 24 h under conditions of 1 mA cm⁻² and 1 mAh cm⁻²: (a) pristine Al; (b) Al/MOF-C. (c,d) Nitrogen adsorption/desorption isotherms and pore size distribution of MOF-C. (e) Structural snapshots of the interfaces between pristine Al and Al/MOF-C in the ionic liquid through MD simulations. (f) Density profiles of AlCl₄⁻ and EMI⁺ for Al/MOF-C surfaces. (g) Differential access mechanism diagram of the MOF-C functional protective layer inhibiting Al corrosion.

terms of the position and intensity of the main characteristic peaks, verifying the successful synthesis of MIL-96. The elemental mapping of MIL-96 exhibits the uniform distribution of elements (Al, C, and O) within MIL-96, further confirming its chemical composition and uniformity (Figure 3c). To systematically evaluate the morphological evolution and interfacial coverage of the MIL-96 protective layer on the Al substrate, we performed high-resolution SEM characterization, as shown in Figure 3d–f. These images conclusively

demonstrate the conformal growth of hexagonal truncated bipyramidal MOF crystals across the Al surface, with an average lateral dimension of 20 μm and vertical thickness of 15 μm (Figure 3g,h). After thermal treatment at 600 °C, the resulting MOF-C retains this characteristic morphology, as evidenced by the SEM images and elemental mapping in Figures S14 and S15, showing no significant structural collapse. Furthermore, the XRD pattern of MOF-C (Figure S13) still exhibits the main diffraction peaks of MIL-96, along with an

additional broad peak at $\sim 23^\circ$ corresponding to carbon (200), confirming that MOF-C is obtained through partial carbonization while retaining the primary MIL-96 framework structure.

Electrochemical Performance of the Al/MOF-C Anode

To investigate the stripping/plating behavior of different electrodes, electrochemical tests were conducted on symmetrical cells composed of two identical electrodes. As shown in Figure 4a, the electrochemical performance of symmetrical cells assembled with three different anodes (pristine Al, Al/MOF, and Al/MOF-C) at 1 mA cm^{-2} and 1 mAh cm^{-2} exhibits significant differences. The cycling time of the pure Al symmetrical cell is only 60 h, with an overpotential of 58 mV, indicating poor electrochemical stability. In contrast, the Al/MOF symmetrical cell shows a greatly extended cycling time to 920 h, but the overpotential increases to 100 mV, suggesting an improvement in battery life but still limited electrochemical performance. Most notably, the Al/MOF-C symmetrical cell achieves an unprecedented cycling lifespan exceeding 11,000 h (the longest ever reported for modified Al anodes) with an ultralow overpotential of merely 20 mV. This record-breaking performance unequivocally demonstrates the exceptional electrochemical stability of Al/MOF-C under prolonged cycling conditions. As shown in Figure 4b, the voltage hysteresis profiles at representative cycles (10th, 20th, 200th, and 2500th) exhibit nearly identical polarization characteristics, directly validating the exceptional long-term stability of Al/MOF-C anodes across 3 orders of magnitude in cycle numbers. Figure S16 shows the voltage distribution of the Al/MOF-C symmetrical cell between 10,950 and 11,000 h, during which the Al/MOF-C anode still exhibits stable plating/stripping curves. Obviously, the Al/MOF-C symmetrical cell maintains a stable voltage platform and a low overpotential throughout the entire cycling process, showing excellent cycling stability. In contrast, the performance of the pure Al and Al/MOF cells gradually declines with increasing cycling numbers, especially the pure Al cell, which shows significant voltage fluctuations and an increased overpotential during long-term cycling. Figure 4c presents the contact angle test results of the three electrodes in an ionic liquid (1:1.3 concentration). The contact angle of pure Al is 79.5° , while the contact angles of Al/MOF and Al/MOF-C are 21° and 20° , respectively. The lower contact angles indicate that the Al/MOF and Al/MOF-C anode materials have significantly better wettability in the ionic liquid than pure Al, which helps to improve electrolyte infiltration on the electrode surface and uniform Al plating/stripping. As shown in Figure 4d, the electrochemical impedance spectroscopy (EIS) results of the three symmetrical cells further reveal the differences in the electrochemical performance. The charge-transfer impedance of the Al/MOF-C symmetrical cell is the lowest, indicating higher charge-transfer efficiency and lower resistance. The lower resistance contributes to reducing polarization in electrochemical reactions, thereby improving the overall performance of the battery, consistent with the symmetrical cell results. In addition, the corresponding equivalent circuit and fitting parameters for the three symmetric cells are provided in Figure S17. The circuit model was used to obtain the fitted resistances, where the Al/MOF-C electrode shows the smallest R_s (2.77Ω) and R_{ct} (54.9Ω) compared to pristine Al and Al/MOF electrodes. These results further confirm that the MOF-C protective layer effectively reduces interfacial

resistance and facilitates more efficient charge transfer, consistent with the EIS trends observed in Figure 4d. Figure 4e shows the relationship between the real part of the impedance and the angular frequency for the three symmetrical cells, indicating that the Warburg coefficient of Al/MOF-C is the smallest, which means the largest ion diffusion coefficient and a higher ion diffusion efficiency. It is suggested that the Al/MOF-C material can more effectively transport ions under high current charge–discharge conditions. As shown in Figure 4f, the cycling time of the Al/MOF-C symmetrical cell in this study far exceeds that of other published works,^{40–47} reaching over 11,000 h. Additionally, the overpotential of the Al/MOF-C symmetrical cell is the lowest, further proving its superior electrochemical performance.

Differential Access Mechanisms of Functional Protective Layer Inhibiting Al Corrosion

As shown in Figure 5a,b and S18, after 24 h of cycling at 1 mA cm^{-2} and 1 mAh cm^{-2} , the surface of pure Al exhibited numerous corrosion pits, and some areas even penetrated by corrosion, where Figure S18 shows SEM images of pristine Al and Al/MOF-C electrodes after 24 h of cycling. In contrast, the Al/MOF-C sample showed a smooth Al surface with no apparent signs of corrosion after washing off the MOF-C protective layer with alcohol, indicating that Al/MOF-C possesses excellent anticorrosion properties. Figures S19 and S20 compare the SEM images of pure Al and Al/MOF-C after cycling for 48 h, revealing no corrosion pits on the Al/MOF-C surface. Figure S21 shows the SEM and elemental mapping images of MOF-C after 48 h of cycling, where the modified material still presents a hexagonal truncated bipyramidal particle structure, demonstrating its structural stability during cycling. Figure S22 compares the Tafel plots of symmetrical cells assembled with the pure Al and Al/MOF-C cells, showing that the absolute value of the equilibrium potential and the corrosion current density of the pure Al cell are significantly higher than those of the Al/MOF-C cell, further proving the superior anticorrosion performance of Al/MOF-C. Figure 5c,d presents the Brunauer–Emmett–Teller test results of MOF-C, where the adsorption and desorption curves are completely consistent, indicating that the material is primarily microporous with an average pore diameter of 0.822 nm .⁵³ According to the Bondi model calculations, the diameters of EMI^+ and AlCl_4^- ions are 0.98 and 0.7 nm , respectively (Figure S23). The pore size of as-designed MOF-C falls between these two values, meaning that MOF-C can effectively screen ions and block the passage of EMI^+ ions while allowing the free migration of AlCl_4^- ions, which is expected to effectively alleviate corrosion caused by EMI^+ ions. To mechanistically validate the ion-sieving capability of MOF-C predicted by pore size analysis, we performed molecular dynamics (MD) simulations probing ionic distributions at the Al/electrolyte interface, as shown in Figure 5e. Unlike the pristine Al anode, the MD trajectory analysis reveals a striking spatial segregation: EMI^+ ions remain confined above the MOF-C layer, forming a depletion zone near the Al surface. This exclusion effect directly manifests MOF-C's ability to block corrosive EMI^+ through size-sieving mechanisms. Additionally, AlCl_4^- ions exhibit greater displacement over the same period, demonstrating that the MOF-C layer can accelerate the transport of AlCl_4^- . The density distribution profiles of AlCl_4^- and EMI^+ at the Al metal surface with the MOF-C functional layer are shown in Figure 5f. The distance

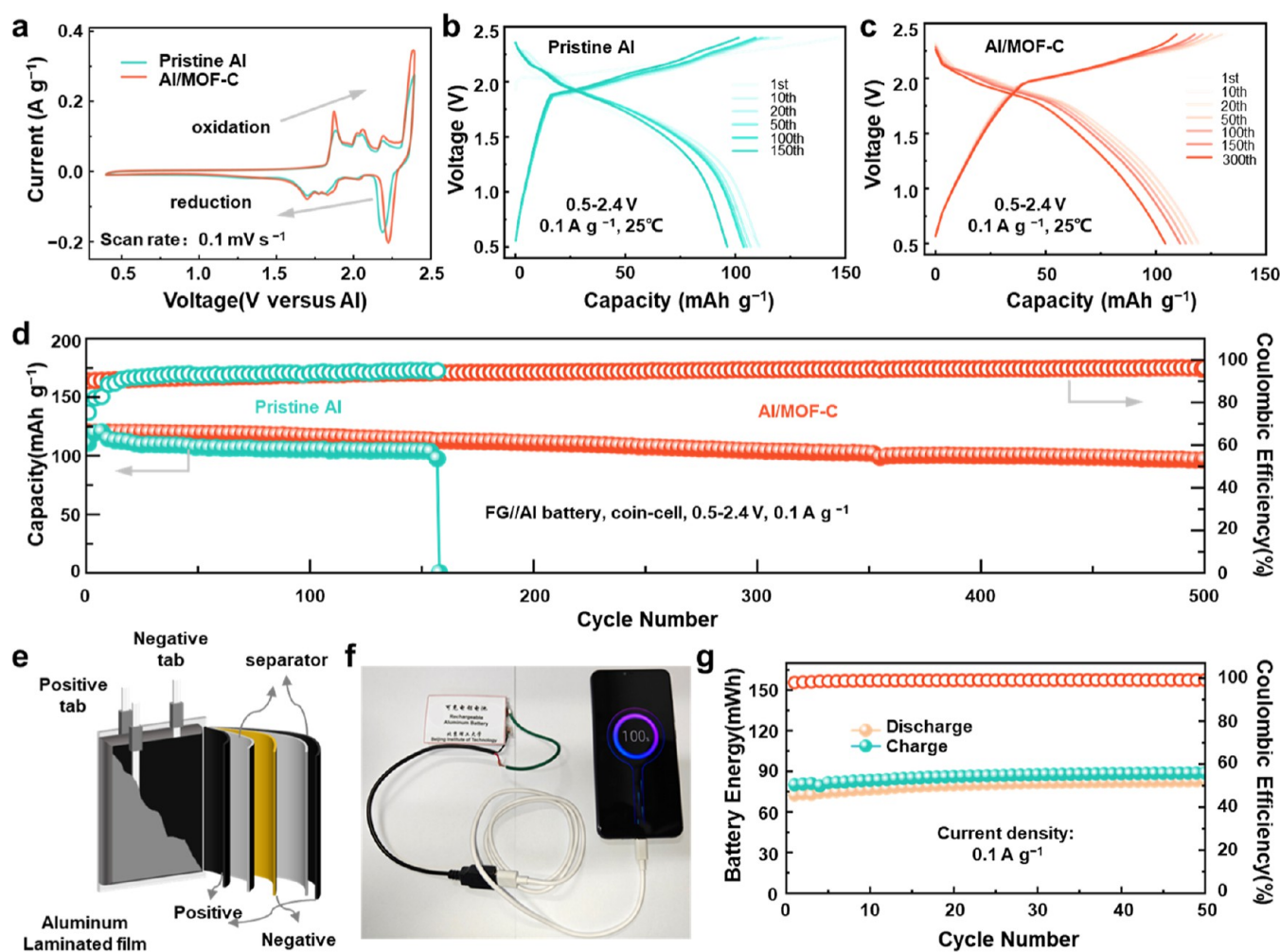


Figure 6. Electrochemical performance of FG//Al full cells: (a) CV curves at 0.1 mV s⁻¹ for pristine Al and Al/MOF-C. (b) Galvanostatic charge/discharge (GCD) profiles of pristine Al at 0.1 A g⁻¹. (c) GCD profiles of Al/MOF-C at 0.1 A g⁻¹. (d) Long-term cycling performance at 0.1 A g⁻¹. (e) Schematic diagram of the FG//Al/MOF-C pouch cell (dimensions: 4.5 × 3 × 0.4 cm). (f) Image of the pouch cell charging a mobile phone. (g) Electrochemical performance of the FG//Al/MOF-C pouch cell.

between the two peaks of EMI⁺ does not exhibit significant changes, suggesting that EMI⁺ hardly penetrates through the MOF-C layer to reach the Al surface. In contrast, the distance between the two peaks of AlCl₄⁻ at 0 and 10 ns increases, indicating that AlCl₄⁻ moves closer to the Al metal surface. Through the comparison of Figures S24–S26, the significantly higher mean-squared displacement (MSD) in the Al/MOF-C system indicates the enhanced mobility of AlCl₄⁻ ions facilitated by the presence of the MOF-C layer. Specifically, Figures S24–S26 present the MSD profiles of AlCl₄⁻, Al₂Cl₇⁻, and EMI⁺ ions at the Al/electrolyte interface, respectively, comparing pristine Al and Al/MOF-C systems. As shown in Figure S27, the radial distribution function $g(r)$ of AlCl₄⁻ ions near the Al/MOF-C interface decreases over time, indicating a transition from localized ion accumulation at 1 ns to a more uniform distribution at 10 ns. This reflects the enhanced ion mobility facilitated by the MOF-C layer's porous structure. These findings are consistent with the electrochemical kinetics results from Figure 5d. Furthermore, galvanostatic intermittent titration technique measurements (Figure S28) reveal that the Al/MOF-C electrode exhibits a diffusion coefficient markedly higher than that of pristine Al. This experimentally confirms the accelerated interfacial ion-transport kinetics predicted by

the MD simulations. In addition, adsorption energy calculations (Figure S29) show that the interaction between AlCl₄⁻ and Al₂Cl₇⁻ is relatively weak, indicating that AlCl₄⁻ is not strongly immobilized by ion pairing and can migrate freely within the MOF-C nanochannels. As shown in Figure 5g, the MOF-C layer imposes spatial constraints due to the size difference between EMI⁺ and AlCl₄⁻ ions; the larger EMI⁺ ions are effectively excluded from penetrating the MOF-C layer and reaching the Al metal surface, thereby preventing Al corrosion. In contrast, the smaller AlCl₄⁻ ions can efficiently pass through the MOF-C pores and reach the Al surface, enabling Al plating and stripping. Furthermore, the MOF-C layer creates a “differential access mechanism”, where the pore size selectively facilitates AlCl₄⁻ ion transport while screening out EMI⁺ ions. This unique mechanism ensures that AlCl₄⁻ ions are the dominant species interacting with the Al metal, which can greatly improve the ion migration efficiency. Furthermore, the porous structure of MOF-C enhances the electrolyte's wettability on the electrode surface, accelerates ion transport, and reduces electrochemical impedance, thereby significantly improving the overall performance of the Al electrode.

Electrochemical Performance of FG//Al Full Cells

Building on the exceptional interfacial stability demonstrated in symmetric cells, we systematically evaluated the practical viability of Al/MOF-C anodes by assembling FG//Al full cells with natural graphite cathodes. As shown in Figure 6a, the cyclic voltammetry (CV) curves of FG//Al batteries using both pure Al and Al/MOF-C anodes exhibit the same redox peaks, corresponding to the insertion and extraction of AlCl_4^- ions. Notably, the FG//Al battery with the Al/MOF-C anode displays a narrower voltage gap between the redox peaks, indicating better redox reversibility of the Al/MOF-C anode in the electrolyte. Figure S30 shows the CV curves of the FG//Al battery with the Al/MOF-C anode at different cycles, where the high overlap of the curves demonstrates the good electrochemical stability of the battery. Figure 6b,c show the charge–discharge curves of FG//Al batteries using pure Al and Al/MOF-C anodes, respectively. The FG//Al battery assembled with pure Al exhibits overcharging in the initial cycle, which is attributed to the presence of an oxide layer on the Al surface, leading to high polarization of the electrode. In contrast, the FG//Al battery assembled with the Al/MOF-C anode does not exhibit overcharging, indicating lower polarization of the Al/MOF-C anode and higher initial Coulombic efficiency of the assembled FG//Al battery. This low polarization performance is crucial for improving battery efficiency and extending its lifespan. Figure 6d displays the specific capacity of FG//Al batteries assembled with the two types of electrodes at a current density of 0.1 A g^{-1} . Although both batteries have an initial specific capacity close to 118 mAh g^{-1} , the FG//Al battery assembled with pure Al fails due to short-circuiting after only 145 cycles. In contrast, the FG//Al battery with the Al/MOF-C anode remains operational and exhibits nearly 100% Coulombic efficiency after over 500 cycles, demonstrating significant stability and durability in long-term cycling. To further investigate the practical applicability of the Al/MOF-C anode, we assembled FG//Al full cells with a higher cathode mass loading of 1.5 mg. As shown in Figure S31, the full cell paired with pristine Al exhibits rapid performance degradation and severe polarization. In contrast, the full cell assembled with the Al/MOF-C anode demonstrates a significantly enhanced rate capability from 0.1 to 1.0 A g^{-1} , highly stable GCD profiles up to 300 cycles, and a superior long-term cycling stability of over 500 cycles with nearly 100% Coulombic efficiency. In addition, a comprehensive comparison of previously reported Al//graphite full cells is presented in Table S1. The table summarizes electrolyte compositions, cathode configurations, current densities, capacities, Coulombic efficiencies, and cycling lifetimes from the representative literature. Compared with these systems, the full cells assembled with our MOF-C-protected Al anode delivered both competitive capacity and markedly improved cycle stability, demonstrating a clear performance advantage. To further validate the practical application of the Al/MOF-C anode, we assembled a pouch cell, as shown in Figure 6e. As shown in Figure 6f and S32, the pouch cell can charge a mobile phone and light up a light-emitting diode lamp, thoroughly showcasing its practical application potential. Figure 6g shows that the battery successfully achieved 50 cycles at a current density of 0.1 A g^{-1} . This work transcends fundamental mechanism exploration, establishing MOF-engineered Al anodes as enablers for high-performance, practically viable RABs ready for scalable deployment.

CONCLUSIONS

In summary, a mechanistic framework for Al anode failure in ionic liquid systems is established, resolving the long-standing ambiguity between dendrite formation and corrosion. By integrating multimodal in situ diagnostics, we demonstrate for the first time that dendrites dominate in the early stages of cycling, but as the cycle time extends, dendrites gradually disappear, and corrosion becomes the critical factor affecting battery performance. This finding pioneeringly underscores the importance of investigating corrosion mechanisms to enhance the long-term stability of Al batteries. Through in situ Raman spectroscopy and theoretical calculations, we precisely identify the role of EMI^+ ions as the primary cause of corrosion and innovatively proposed the in situ construction of a MOF-C functional protective layer on the Al surface to inhibit corrosion. Electrochemical test results show that the Al/MOF-C anode in FG//Al batteries exhibits exceptional electrochemical performance, with significant anticorrosion capability and ultralong cycle stability, achieving a cycling time of over 11,000 h and an ultralow overpotential of only 20 mV. Moreover, the Al/MOF-C anode demonstrates strong practical potential in real-world applications. Coupled with natural graphite, the assembled FG//Al batteries achieve a high specific capacity of 120 mAh g^{-1} and successfully cycle for 500 cycles, effectively improving battery efficiency and extending lifespan. This study not only deepens the understanding of the interfacial evolution and corrosion mechanisms of Al metal in the $[\text{EMI}]\text{Cl}-\text{AlCl}_3$ ionic liquid electrolyte system but also provides innovative design principles and technical pathways for developing efficient and durable Al-based energy storage materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c22732>.

Full experimental details, mechanistic studies, computational studies, and characterization of cells (PDF)

Surface morphology changes during the charge–discharge process of the Al symmetrical battery (MP4)

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Notes

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